

Lab Manual — Week 9

Concurrency & Distributed Systems — Lebanese University, Faculty of Engineering

Topic: Python GIL and Runtime Limits

Learning Objectives

1. Explain why CPU-bound CPython threads do not scale across cores.
2. Compare threads and multiprocessing for CPU-bound work.
3. Capture visual CPU evidence with Task Manager, `htop`, or `py-spy`.
4. Contrast Python with Go goroutines and Erlang actors.
5. Connect Knight Capital to timing, deployment, and operational controls.

Lab Folder

```
week9-lab/week9-day1-python-gil-runtime-limits
```

Use:

```
.\run-tests.ps1
.\student-check.ps1
.\run-all-demos.ps1
```

Part 1 — Threads vs Processes

Open:

```
src/week9/part1_threads_vs_processes/exercises.py
```

Task: Explain CPU-bound Python threads, multiprocessing, and multiprocessing cost.

Part 2 — Visual Proof

Open:

```
VISUAL_PROOF_GUIDE.md
src/week9/part2_visual_proof/exercises.py
```

Task: Describe a concrete visual proof plan.

Part 3 — Language Contrast

Open:

```
LANGUAGE_CONTRAST.md
src/week9/part3_language_contrast/exercises.py
```

Task: Contrast Python GIL, Go goroutines/channels, and Erlang actors/message passing.

Part 4 — Knight Capital Discussion

Open:

`case-studies/W9_Knight-Capital-2012.md`
`src/week9/part4_case_study/exercises.py`

Task: Connect deployment timing, stale code, and operational controls to concurrency risk.

Submission Checklist

- ☐ `student-check.ps1` passes.
- ☐ Visual proof note is complete.
- ☐ Language contrast is complete.
- ☐ Knight Capital discussion points are complete.

Bridge to Week 10

Week 10 starts distributed systems. There is no shared memory across the network. Queues become sockets, and failure becomes partial and harder to observe.